

I Year – I Semester

Course Code	PHYSICS FOR CIVIL ENGINEERS (AI Integrated)	L	T	Pr	P	Credits
26APH01T		3	0	0	0	3

Prerequisites: Engineering Mathematics, Basics of Physics, Newton's laws, Ray optics.

Course Objectives

1. To help students understand the concepts of interference, diffraction, and polarization and their applications in optical measurements, surveying, and civil engineering structures, with basic AI-assisted analysis techniques.
2. To introduce crystallography and X-ray diffraction methods for studying crystal structures and construction materials with simple AI-assisted material characterization concepts.
3. To provide knowledge on quantum mechanics and quantum computing concepts, along with basic AI-assisted quantum simulation and secure communication applications.
4. To familiarize students with oscillations, vibration control, and acoustics of buildings used in earthquake-resistant and smart structures with simple AI-assisted monitoring techniques.
5. To impart knowledge of ultrasonics and non-destructive testing methods for quality inspection and defect detection in civil engineering structures with basic AI-assisted inspection concepts.

Course Content

UNIT I: Applied Wave Optics

Core Content:

Interference: Principle of superposition and coherence, Interference, Types of interference, Interference in thin films (reflection geometry) and Newton's rings. Applications: Determination of wavelength and refractive index.

Diffraction and Optical Surveying: Introduction to diffraction, Fresnel and Fraunhofer diffraction. Fraunhofer diffraction due to single, double and N-slits (Diffraction Grating), Dispersive power and Resolving power of grating, Applications: Measurement of distances, angles and surface alignment in civil engineering (qualitative).

Polarization: Types of polarization, Nicol's prism, Half-wave and Quarter-wave plates.

AI Integration:

AI-Assisted Flatness Detection Using Interference Fringes, AI-Based Mineral Analysis and Degradation Detection in Masonry and Concrete Using Diffraction.

Free/Open-Source AI Tools: Python (NumPy, SciPy, Matplotlib and scikit-learn).

UNIT II: Crystallography and X-Ray Diffraction

Core Content:

Crystallography: Space lattice, Basis, Unit Cell and lattice parameters – Crystal systems, Bravais Lattices — Coordination number - Packing fraction of SC, BCC & FCC - Miller indices – Separation between successive (h k l) planes.

X-ray diffraction: Bragg's law - X-ray Diffractometer – Crystal structure determination by Laue's method. Application: Checking the quality of construction materials.

AI Integration:

AI-driven XRD for Real-time corrosion assessment in reinforced concrete and identifying micro-cracks in ageing infrastructure.

Free/Open-Source AI Tools: Google Colab, Python, NumPy, SciPy and Scikit-learn/ TensorFlow.

UNIT III: Introduction to Quantum Mechanics & Quantum Computing**Core Content:**

Introduction to quantum mechanics, de-Broglie concept (matter waves), Heisenberg's Uncertainty Principle, Wave function, Postulates of Quantum Mechanics, Schrödinger's time-independent wave equation, Particle in a one-dimensional potential well, Quantum tunnelling (basic idea).

Introduction to Quantum Computing: Classical bits vs qubits. Concept of superposition and Entanglement, Basic Quantum Operations: Hadamard gate (idea of superposition), Pauli-X gate, Basic idea of quantum measurement, Applications: Basic idea of quantum cryptography and secure communication.

AI Integration: AI-assisted quantum circuit simulation using IBM Qiskit (free). Python-based visualization of wave functions and quantum probability densities.

Free/Open-Source AI Tools: Google Colab, Python and Scikit-learn.

UNIT IV: Oscillations and Acoustics of Buildings**Core Content:**

Introduction to oscillations, Oscillations of springs, Series and parallel combinations of springs (quantitative), Stiffness factor and its physical significance. Engineering applications of springs in Base Isolation Systems for earthquake-resistant buildings. Damping and Resonance: Theory of damped oscillations (Over, Under and Critical damping- all qualitative); Forced oscillations (qualitative) and Resonance. Sharpness of resonance and Quality factor (Q). Applications: Tuned Mass Dampers (TMDs) for seismic resilience and wind-induced vibration control in skyscrapers (qualitative).

Acoustics of Buildings: Introduction to architectural acoustics; Reverberation and reverberation time, Sabine's formula (qualitative). Factors affecting architectural acoustics and requirements for acoustically optimized halls. Applications - Active Noise Cancellation (ANC) systems for smart urban facades (Qualitative).

AI Integration: AI-Assisted Strategies for noise mitigation in Green Buildings. Use of Acoustic Simulation AI for auditorium design.

Free/Open-Source AI Tools: Google Colab, Python and Scikit-learn.

UNIT V: Ultrasonics and Non-Destructive Testing**Core Content:**

Introduction to ultrasonics: Production of ultrasonics: Piezoelectric method and Magnetostriction method. Detection of ultrasonics: Kundt's tube and Quartz crystal methods. Role of Ultrasonic Pulse Velocity (UPV) testing in assessing concrete quality, uniformity and depth of cracks in structural elements.

Non-Destructive Testing (NDT): Introduction, need and advantages in civil engineering structures. Basic inspection methods: Visual inspection and AI-based drone inspections for inaccessible structures (qualitative). Liquid penetrant testing: basic principle for detecting surface defects in steel. Application: Electromagnetic inspection: eddy current testing (to detect surface cracks, corrosion and defects in conductive materials). (basic idea)

AI Integration: AI-Powered Defect Detection in Structural Steel, Weld Joints and Corrosion in Reinforced Concrete (RCC)

Free/Open-Source AI Tools: Google Colab, Python and Scikit-learn.

Course Outcomes (COs)

After completion of the course, the student will be able to

CO	Course Outcomes	Bloom's Level
CO1	Analyze the principles of interference, diffraction, and polarization and apply basic AI-assisted optical techniques for surveying, surface analysis, and civil engineering applications.	L4 (Analyze)
CO2	Examine crystal structures and interpret X-ray diffraction patterns for material characterization using basic AI-assisted analysis techniques in civil engineering materials.	L4 (Analyze)
CO3	Apply the concepts of quantum mechanics and quantum computing to explain quantum-based communication systems with basic AI-assisted quantum simulations.	L3 (Apply)
CO4	Analyze oscillatory systems, vibration control mechanisms, and architectural acoustics for civil engineering structures using basic AI-assisted monitoring and acoustic analysis methods.	L4 (Analyze)
CO5	Evaluate ultrasonic and non-destructive testing techniques for identifying defects and assessing the quality of structural materials using basic AI-assisted defect detection approaches.	L5 (Apply)

CO – PO Articulation Matrix

Scale: 3 – Strong, 2 – Moderate, 1 – Slight

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	2	2	3	–	–	–	–	–	–
CO2	3	3	2	2	3	–	–	–	–	–	–
CO3	3	3	2	2	2	–	–	–	–	–	–
CO4	3	3	3	2	3	–	–	–	–	–	–
CO5	3	3	2	3	3	–	–	–	–	–	–

Textbooks

1. Avadhanulu, Kshirsagar & Arun Murthy, A Textbook of Engineering Physics, S. Chand, 2019.
2. D. K. Bhattacharya & Poonam Tandon, Engineering Physics, Oxford Press, 2015.

Reference Books

1. Engineering Physics — B. K. Pandey & S. Chaturvedi — Cengage Learning, 2021
2. Engineering Physics — Shatendra Sharma & Jyotsna Sharma — Pearson, 2018
3. Engineering Physics — Sanjay D. Jain, D. Sahasrabudhe & Girish — Universities Press, 2010
4. Engineering Physics — M. R. Srinivasan — New Age International, 2009
5. Solid State Physics — S. O. Pillai — New Age International
6. Modern Physics — R. Murugesan & Kiruthiga Sivaprasath — S. Chand

Web Resources & NPTEL Links

- NPTEL – nptel.ac.in
- MIT OpenCourseWare – ocw.mit.edu
- HyperPhysics – hyperphysics.phy-astr.gsu.edu
- Feynman Lectures – feynmanlectures.caltech.edu
- PhET Simulations – phet.colorado.edu

Recommended Free & Open-Source AI/Computational Tools

- Python (NumPy, SciPy, OpenCV, Matplotlib) – FREE on Google Colab for optics and EM simulations
- IBM Qiskit (free, open-source) – For quantum computing and quantum cryptography simulations
- scikit-learn (free, open-source) – For semiconductor data analysis and signal classification
- PhET Interactive Simulations (free) – For wave optics, semiconductors and quantum phenomena

I Year I Semester

Course Code	Physics for Civil Engineers Lab (AI integrated)	L	T	Pr	P	Credits
26APH01P		0	0	0	3	1.5

Prerequisites: Basic Physics and Mathematics (Class XII level); no concurrent prerequisite.

Course Objectives

1. To measure refractive indices, light dispersion, surface curvature, and thin-film thickness using interference and prism spectroscopy techniques.
2. To characterize spectral emission lines, laser diffraction patterns, and microscopic particle dimensions using transmission gratings.
3. To investigate quantum photoelectric thresholds and evaluate static and dynamic elastic properties of mechanical materials.
4. To analyze multi-degree-of-freedom oscillatory modes, spring system dynamics, and liquid refractometry principles.
5. To evaluate acoustic propagation speed, standing wave resonance, and sound wave velocities in solid wires and liquid media.

List of Experiments (Any ten experiments need to be performed)**Part A – Regular Experiments**

1. Determination of the dispersive power of the prism.
2. Determination of the thickness of a thin object by the wedge method
3. Determination of the radius of curvature of the lens by Newton's rings
4. Determination of wavelengths of various colours of mercury spectrum using diffraction grating in the normal incidence method
5. Determination of wavelength of a LASER using a diffraction grating
6. Determination of particle size using a diode laser
7. Estimation of Planck's constant using the photoelectric effect
8. Determination of spring constant by Hooke's law
9. Melde's experiment: Determination of the frequency of the tuning fork
10. Sonometer: Verification of the three laws of stretched strings
11. Determination of frequency of normal modes of the Coupled Oscillator
12. Study of the Refractive Index of Materials using a hollow prism used in surveying
13. Study of oscillations of a mass under different combinations of springs (Loaded Springs)
14. Determination of ultrasonic velocity in liquid (Acoustic grating)
15. Determination of Rigidity modulus of material of a wire-dynamic method (Torsional pendulum)

Part B – AI Integration Lab Exercises

Any **three** of the following AI exercises must be completed using Python / Google Colab or appropriate AI/LLM tools.

1. **Spring Constant Data Automation (Python / Matplotlib):** input load vs. extension data from the Hooke's law experiment; fit a straight line ($F = kx$) using simple linear regression; extract the spring constant k from the slope; plot the fitted line over the experimental points and compute percentage error against the manually calculated value.
2. **Ultrasonic Pulse Data Visualization and Defect Interpretation (Python / Google Colab):** Use sample ultrasonic pulse travel-time data collected from concrete specimens (or instructor-provided datasets). Plot pulse velocity against specimen condition, compare normal and defective samples, and qualitatively identify possible cracks or internal defects using simple graphical analysis.
3. **Prism / Surveying-Refractometer Data Analysis (Python / Matplotlib):** input angle-of-incidence vs. angle-of-deviation data from the prism or hollow-prism (surveying) experiment; plot the deviation curve; locate the angle of minimum deviation; compute the refractive index and dispersive power; compare with the manually calculated result.

Free / Open-Source AI Tools

Python (NumPy, SciPy, Matplotlib) – Google Colab (free); scikit-learn (free) – linear regression and curve fitting; PhET Interactive Simulations (free) – wave optics and mechanical oscillations; VLabs IIT (free) – online physics and surveying-instrument simulations

Course Outcomes (COs)

After successful completion of this lab course, the student will be able to:

CO	Course Outcomes	Bloom's Taxonomy Level
CO1	Compute refractive indices, dispersive power, surface curvature, and thin-film thickness by operating prism spectrometers and interference setups.	L3 (Apply)
CO2	Analyze spectral emission lines, laser diffraction patterns, and particle sizes using transmission diffraction gratings.	L3–L4 (Apply, Analyse)
CO3	Evaluate quantum photoelectric energy thresholds alongside static and dynamic elastic properties of mechanical materials.	L4 (Analyse)
CO4	Analyze multi-degree-of-freedom oscillatory modes, spring system dynamics, and liquid refractive properties through harmonic and refractometric setups.	L4–L5 (Analyse, Evaluate)
CO5	Evaluate acoustic propagation speeds, standing wave resonance, and sound wave velocities across solid wires and liquid media.	L5–L6 (Evaluate, Create)

CO – PO Articulation Matrix (NBA)*Scale: 3 – Strong | 2 – Moderate | 1 – Slight | – No Correlation*

CO	Bloom's Lvl	PO1 Eng. Know.	PO2 Prob. Analysis	PO3 Design/Dev.	PO4 Investigation	PO5 Modern Tools	PO6 Eng. & Society	PO7 Env. & Sust.	PO8 Ethics	PO9 Team Work	PO10 Comm.	PO11 Prj Mgmt
CO1	L3	3	2	2	3	2	–	–	–	–	–	–
CO2	L3–L4	3	3	3	3	3	–	–	–	–	–	–
CO3	L4	3	3	2	3	2	–	–	–	–	–	–
CO4	L4–L5	3	3	3	3	3	–	–	–	–	–	–
CO5	L5–L6	2	3	3	3	3	–	–	–	–	–	–

Recommended Free & Open-Source AI / Computational Tools

Python (NumPy, SciPy, Matplotlib) on Google Colab (free) – spring-constant regression, sonometer law verification, prism/surveying deviation-curve analysis

scikit-learn (free, open-source) – linear regression and curve fitting for lab data

PhET Interactive Simulations (free) – wave optics, oscillations, and quantum phenomena: phet.colorado.edu

Virtual Labs – IIT (free): https://iitb.vlabs.co.in/discipline.html?discipline=Physical_Sciences – online simulation of optics and mechanics experiments

HyperPhysics (free) – concept reference for optics, mechanics, and quantum theory: hyperphysics.phy-astr.gsu.edu

LibreOffice Calc (free, open-source) – standardised data recording templates for all experiments

Note: Lab record to be submitted for every experiment at the next session. AI exercises to be submitted as Google Colab.ipynb notebooks.

Reference Books

1. M. N. Avadhanulu & P. G. Kshirsagar, A Textbook of Engineering Physics, S. Chand & Company, 2019.
2. H. K. Malik & A. K. Singh, Engineering Physics, Tata McGraw-Hill, 2010.
3. R. K. Gaur & S. L. Gupta, Engineering Physics, Dhanpat Rai Publications, 2012, 8th Edition.
4. B. K. Pandey & S. Chaturvedi, Engineering Physics, Cengage Learning, 2012.